

NPTEL Online Certification Courses
Indian Institute of Technology Kharagpur



Introduction to
Internet of Things
Assignment-Week 9

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

OpenStack was launched in 2010 as a joint project between Rackspace Hosting and which other organization?

a. Google

b. Microsoft

c. NASA

d. IBM

Correct Answer: c. NASA

Detailed Solution: OpenStack, a software used to create cloud infrastructure, was launched as a joint project of Rackspace Hosting and NASA in 2010, and is open source.

See Lecture 41 (Cloud Computing – Practical, **slide 3**)

QUESTION 2:

Which OpenStack component provides the dashboard (GUI) that gives an overview of the other components?

a. Nova

b. Keystone

c. Horizon

d. Swift

Correct Answer: c. Horizon

Detailed Solution: Horizon is the dashboard component of OpenStack — it provides the GUI of the software and gives an overview of the other components.

See Lecture 41 (Cloud Computing – Practical, **slide 5**)

QUESTION 3:

Which OpenStack component is the compute service where instances are launched?

a. Nova

b. Glance

c. Cinder

d. Neutron

Correct Answer: a. Nova

Detailed Solution: Nova is the compute service in OpenStack — it is where you launch your instances. Glance is the image service, Cinder is block storage, and Neutron provides networking.

See Lecture 41 (Cloud Computing – Practical, **slide 6**)

QUESTION 4:

Which script/tool is used in the lecture to install OpenStack, requiring the local.conf file to be configured with passwords and host IP before running?

a. stack.sh via Devstack

b. docker-compose

c. kubeadm

d. ansible-playbook

Correct Answer: a. stack.sh via Devstack

Detailed Solution: OpenStack is installed using Devstack: after cloning the devstack repository and configuring local.conf with ADMIN_PASSWORD, DATABASE_PASSWORD, RABBIT_PASSWORD, SERVICE_PASSWORD, and HOST_IP, the stack.sh file is run to install it.

See Lecture 41 (Cloud Computing – Practical, **slides 9–10**)

QUESTION 5:

According to the lecture, what does sensor-cloud fundamentally provide, beyond merely integrating sensors with cloud computing?

a. A physical replacement for all sensor hardware

b. Virtualization of sensor nodes, offered on a pay-per-use basis

c. A way to eliminate the need for a sink node

d. Free unlimited sensor deployment

Correct Answer: b. Virtualization of sensor nodes, offered on a pay-per-use basis

Detailed Solution: Sensor-cloud is not mere integration of sensors and cloud computing or 'dumping sensor data into the cloud' — it introduces the concept of virtualization of the sensor node, offered on a pay-per-use basis, acting as a stratum between sensor nodes and end-users.

See Lecture 42 (Sensor-Cloud – Part I: Sensor-as-a-Service, **slide 10**)

QUESTION 6:

In a traditional Wireless Sensor Network (WSN), who is typically responsible for redeployment of the sensor nodes?

a. The Sensor-Cloud Service Provider (SCSP)

b. The end-user of the sensor-cloud

c. The WSN-user

d. A third-party auditor

Correct Answer: c. The WSN-user

Detailed Solution: In a traditional WSN, the WSN-user is responsible for ownership, deployment, redeployment, maintenance, overhead, and usage. In sensor-cloud, redeployment and maintenance shift to the Sensor-Cloud Service Provider (SCSP), while usage shifts to the end-user.

See Lecture 42 (Sensor-Cloud – Part I: Sensor-as-a-Service, **slide 12**)

QUESTION 7:

In the sensor-cloud model, which actor purchases physical sensor devices and lends them to the sensor-cloud, playing a role from a business perspective?

a. End-user

b. Sensor-owner

c. Sensor-Cloud Service Provider (SCSP)

d. WSN-user

Correct Answer: b. Sensor-owner

Detailed Solution: The Sensor-owner purchases physical sensor devices deployed over different geographical locations and lends these devices to the sensor-cloud, acting as a business actor.

See Lecture 42 (Sensor-Cloud – Part I: Sensor-as-a-Service, **slide 13**)

QUESTION 8:

Which of the following is listed as a major limitation of traditional WSNs that motivates the sensor-cloud model?

a. Unlimited scalability at zero cost

b. Difficulty adapting to a change of requirement (e.g., moving from agriculture to smart home use)

c. Excessive availability of skilled vendors

d. No need for maintenance

Correct Answer: b. Difficulty adapting to a change of requirement (e.g., moving from agriculture to smart home use)

Detailed Solution: Limitations of WSNs include procurement issues (price, right vendor, sensor types), deployment and maintenance overheads, and difficulty adapting to a change of requirement — e.g., a WSN built for agriculture cannot easily be repurposed for a smart home.

See Lecture 42 (Sensor-Cloud – Part I: Sensor-as-a-Service, **slide 9**)

QUESTION 9:

In the Optimal Composition of Virtual Sensors, what does CoV-I refer to?

a. Composition of virtual sensors spanning multiple geographic regions

b. Composition of virtual sensors using only homogeneous nodes within the same geographic region

c. A pricing model for sensor-cloud

d. A caching mechanism for sensor data

Correct Answer: b. Composition of virtual sensors using only homogeneous nodes within the same geographic region

Detailed Solution: CoV-I refers to the formation of a Virtual Sensor (VS) using homogeneous sensor nodes within the same geographical boundary, while CoV-II spans multiple regions using heterogeneous nodes to form a Virtual Sensor Group (VSG).

See Lecture 43 (Sensor-Cloud – Part II: Sensor-as-a-Service, **slide 6**)

QUESTION 10:

In the sensor-cloud caching mechanism, which cache handles requests directly from the end-user and decides whether to serve data directly or re-cache from another source?

a. External Cache (EC)

b. Sensor Cache (SC)

c. Distributed Cache (DC)

d. Internal Cache (IC)

Correct Answer: d. Internal Cache (IC)

Detailed Solution: The Internal Cache (IC) handles requests from the end-user and takes the decision on whether the data should be provided directly or whether it is required to re-cache the data from the External Cache (EC).

See Lecture 43 (Sensor-Cloud – Part II: Sensor-as-a-Service, **slide 13**)

QUESTION 11:

Why is dynamic and adaptive data caching useful in sensor-cloud, according to the lecture?

a. Because environmental conditions always change extremely fast

b. Because it removes the need for a sensor-owner

c. Because caching eliminates the need for virtualization

d. Because unnecessary sensing during slow environmental change wastes energy

Correct Answer: d. Because unnecessary sensing during slow environmental change wastes energy

Detailed Solution: In many cases, environmental conditions change quite slowly, so the sensed data of physical sensors remains largely unaltered; unnecessary sensing under such conditions causes needless energy consumption, motivating adaptive caching.

See Lecture 43 (Sensor-Cloud – Part II: Sensor-as-a-Service, **slide 12**)

QUESTION 12:

Which company is credited with coining the term 'fog computing'?

a. Google

b. Amazon

c. CISCO

d. IBM

Correct Answer: c. CISCO

Detailed Solution: Fog computing, or 'fogging,' is a term coined by CISCO. The idea is to extend the cloud nearer to IoT devices by introducing an intermediate layer between the cloud and the devices.

See Lecture 44 (Fog Computing – Part I, **slide 2**)

QUESTION 13:

Which of the following is listed as a key issue with the traditional cloud model that motivates fog computing?

a. Excess bandwidth availability

b. Volume, latency, and bandwidth limitations of the cloud

c. Too much local processing power at the device level

d. Absence of any IoT devices

Correct Answer: b. Volume, latency, and bandwidth limitations of the cloud

Detailed Solution: The current cloud model faces issues of Volume (too much data to process/store), Latency (round-trip time is too slow for time-sensitive data), and Bandwidth (huge traffic generated by billions of devices consumes most available bandwidth).

See Lecture 44 (Fog Computing – Part I, **slide 6**)

QUESTION 14:

Where is the fog layer positioned in the fog computing architecture?

a. Above the cloud layer

b. Between the cloud and the IoT devices

c. Inside each IoT device only

d. There is no fixed position; it replaces the cloud entirely

Correct Answer: b. Between the cloud and the IoT devices

Detailed Solution: Fog is a layer between the cloud and IoT devices; many fog nodes can be present, and sensor data are processed in the fog before being sent to the cloud, reducing latency and saving bandwidth and cloud storage.

See Lecture 45 (Fog Computing – Part II, **slide 2**)

QUESTION 15:

Which of the following is NOT listed as a required characteristic of a fog node?

a. Storage (transient)

b. Computing facility for quick decisions

c. Network connectivity

d. Permanent, unlimited archival storage

Correct Answer: d. Permanent, unlimited archival storage

Detailed Solution: Fog nodes need transient storage, computing facility (to process data and take quick decisions), and network connectivity to connect with IoT devices, other fog nodes, and the cloud — not permanent unlimited archival storage, which is the cloud's role.

See Lecture 45 (Fog Computing – Part II, **slide 4**)

**********END**********